

Smart Metro Ticket Generation System

Phase 1.2 | Brainstorming

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks



This document captures the **Brainstorming** stage of Phase 1.2 for the Smart Metro Ticket Generation System — a ServiceNow-based application for metro ticket booking, passenger management, fare calculation and ticket generation. The goal of this session is to generate a wide, judgement-free pool of ideas before narrowing down to a problem statement and solution design in the phases that follow. Brainstorming sits at the very start of the ideathon journey — the quality and range of ideas captured here directly shapes how strong the final solution can be.

Session Overview

Item	Details
Objective	Generate ideas to modernize metro ticketing through automation on ServiceNow
Scope	Ticket booking, passenger management, fare calculation, ticket generation
Phase	Phase 1.2 — Brainstorming (Ideation stage of the ideathon roadmap)
Participants	Cross-functional ideathon team (development, design, business analysis)
Duration	Approx. 2 hours, time-boxed
Output	Consolidated idea pool, grouped by theme, feeding into Problem Statement

Guiding Principles for the Session

Principle	What It Means in Practice
Defer judgement	No idea is criticized or ranked during the session itself
Encourage wild ideas	Bold, even impractical ideas are welcomed — they often spark practical ones
Go for quantity	More ideas first; quality filtering happens in a later phase

02 Idea Pool Captured

Every idea raised during the session, regardless of feasibility, was logged first and evaluated later — keeping the brainstorming phase judgement-free. The pool below spans the full ticketing lifecycle, from the moment a commuter opens the app to the moment operations staff review the day's ridership.

Idea	Description	Category
Self-service Booking Portal	Web/mobile portal on ServiceNow Service Portal for instant ticket booking	Booking
Auto Fare Engine	Rule-based fare calculation using distance/zone tables	Fare
Digital Passenger Profile	Central record storing passenger history, passes, preferences	Passenger Mgmt
QR / Digital Ticket Generation	Auto-generate scannable QR ticket as a record on confirmation	Ticketing
Smart Notifications	SMS/Email/Push alerts for booking confirmation, reminders, refunds	Engagement
Admin Dashboard & Reports	Analytics dashboards for ridership, revenue, peak-hour load	Analytics
Chatbot Assistance	Virtual Agent for FAQs, booking help, complaint logging	Support
Group / Family Booking	Book multiple tickets under one transaction with shared fare split	Booking
Auto Refund Engine	Automatic refund trigger for cancelled or delayed journeys	Fare
Contactless Gate Validation	QR ticket scanned directly at entry/exit gates, no physical check	Ticketing
Frequent Traveler Pass	Auto-suggest monthly/weekly pass based on travel pattern	Passenger Mgmt
Multilingual Interface	Booking portal available in regional languages for wider accessibility	Engagement
Peak-Hour Demand Alerts	Warns commuters of crowded trains and suggests off-peak slots	Analytics
Lost Ticket Recovery	Self-service flow to reissue a digital ticket without staff involvement	Support
Accessibility Mode	Larger text, screen-reader support for elderly and differently-abled commuters	Engagement
Staff Duty Roster Sync	Links counter/gate staff scheduling with expected passenger volume	Passenger Mgmt

Note: Ideas above are unfiltered raw output of the brainstorming stage; prioritization and feasibility scoring happen in later phases.

03 Idea Categorization & Next Steps

Ideas from the pool were grouped into themes to make it easier to spot overlaps and decide what to carry forward into the Problem Statement stage. Grouping also makes gaps visible — a theme with very few ideas may need a second look before the team moves on.

Theme	Ideas Grouped Under It
Booking Experience	Self-service Booking Portal, Group/Family Booking, Multilingual Interface
Fare & Payments	Auto Fare Engine, Auto Refund Engine
Ticket Lifecycle	QR/Digital Ticket Generation, Contactless Gate Validation, Lost Ticket Recovery
Passenger Relationship	Digital Passenger Profile, Frequent Traveler Pass, Staff Duty Roster Sync
Communication & Support	Smart Notifications, Chatbot Assistance, Accessibility Mode
Operations & Insight	Admin Dashboard & Reports, Peak-Hour Demand Alerts

Key Takeaways

- Booking, fare and ticketing form the essential core — every other idea supports these three
- Automation is the common thread across nearly every idea raised
- Passenger-facing convenience (notifications, multilingual support, accessibility) came up as often as backend logic, showing the team instinctively balanced both sides of the system
- A few ideas (staff roster sync, peak-hour alerts) hint at operational value beyond the commuter, worth keeping even if they aren't part of the first MVP

Next Steps

- Converge these ideas into a single, crisp problem statement (Phase 1.3)
- Validate top ideas against a commuter empathy map before solution design
- Score shortlisted ideas against the evaluation criteria above
- Carry forward high-value, low-effort ideas as candidates for MVP scope

Closing Note

This brainstorming record is intentionally kept broad. Nothing here is final — its purpose is to give the next phase, Challenge Definition, a rich and honest starting pool to converge from, rather than a narrow set of pre-decided features.